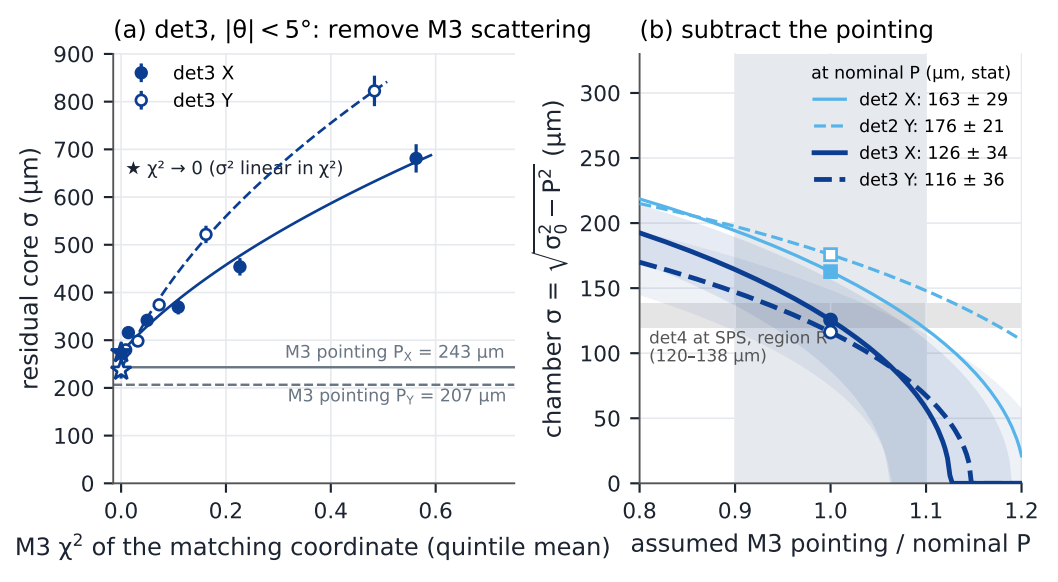
(a) det3, |θ| < 5°: remove M3 scattering
(b) subtract the pointing
det3 X
det3 Y
★ χ² → 0 (σ² linear in χ²)
M3 pointing P_X = 243 μm
M3 pointing P_Y = 207 μm
residual core σ (μm)
M3 χ² of the matching coordinate (quintile mean)
chamber σ = √(σ₀² − P²) (μm)
assumed M3 pointing / nominal P
at nominal P (μm, stat)
det2 X: 163 ± 29
det2 Y: 176 ± 21
det3 X: 126 ± 34
det3 Y: 116 ± 36
det4 at SPS, region R
(120–138 μm)